

Adaptive Digital Phantom

A Developmental Reproducible Research Framework for Patient-Adaptable CT,
Vascular Flow, and Radiotherapy Digital Phantom Development

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Abstract

The Adaptive Digital Phantom project is a modular research framework for constructing anatomically adaptable digital human phantoms from CT-derived anatomy, segmentation labels, material-density maps, vascular graph models, flow simulations, radiotherapy QA workflows, and three-dimensional renderable outputs. The present implementation focuses on a torso and abdominal digital phantom rather than a physical 3D-printed phantom. It supports patient-profile adaptation using BMI, waist circumference, and height; ingestion of CT scans with optional organ and vessel segmentations; vascular graph generation and voxelization; first-pass steady and pulsatile one-dimensional flow modeling; radiotherapy planning and QA scaffolding; and automated reporting. This white paper describes how the model was developed, how it works, its major software modules, practical use cases, built-in packages, current readiness, and limitations. The framework is intended for research and engineering development only. It is not a clinical device, diagnostic system, treatment-planning system, or validated patient-specific digital twin.

Repository: https://github.com/kaiwoodger/Adaptive_Digital_Phantom-

1 Introduction

Digital anthropomorphic phantoms are used in imaging, radiation therapy, medical physics, device evaluation, and computational simulation. Traditional digital phantoms often provide fixed anatomy, limited patient-specific adaptation, or isolated modeling of only one domain such as imaging, radiation dose, or flow. The Adaptive Digital Phantom project was developed to explore a more integrated approach: a reproducible software framework in which anatomy, material properties, vascular networks, flow states, radiotherapy QA, and renderable geometry can be generated and evaluated together.

The project began with a broad aim: to design a lifelike full-body phantom that could function as a digital twin for testing. The initial ambition included real-density tissue values, blood-flow dynamics, correct anatomical regions, CAD compatibility, and realistic medical imaging behavior. During development, the scope was narrowed to a more achievable first product: a torso and abdominal digital phantom focused on CT anatomy, radiotherapy QA, and vascular flow. The work later shifted away from immediate physical 3D printing and toward a purely digital, reproducible, patient-adaptable framework.

2 Project Objectives

The objective of the framework is to create a digital phantom platform that can:

- Represent anatomy using CT-derived organ labels and material-density maps.
- Adapt body habitus using BMI, waist circumference, and height.

- Accept user CT scans and optional organ or vessel segmentations.
- Generate or register vascular networks into the phantom volume.
- Convert vascular geometry into solver-facing arterial and venous flow domains.
- Simulate first-pass steady and pulsatile vascular flow.
- Produce radiotherapy QA packages and compare static versus flow-modulated dose states.
- Export renderable 3D meshes, preview images, reports, manifests, and release packages.
- Remain reproducible from a clean source checkout without committing restricted medical data.

3 System Overview

The software is organized as a Python command-line toolkit called **phantom-twin**. Each command performs one stage of the phantom workflow. The repository separates source code from medical data and generated outputs, allowing the codebase to be published while local users stage their own CT, CTA, CTV, organ segmentation, and vessel segmentation data according to each dataset’s license.

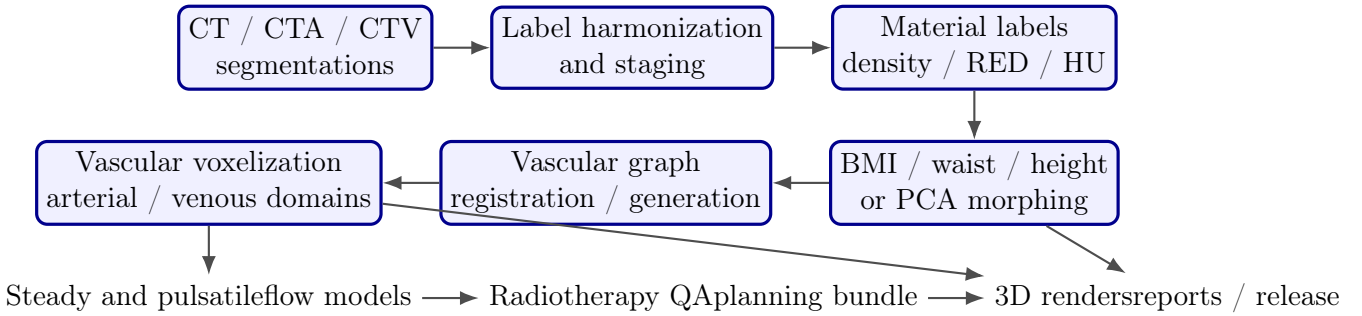
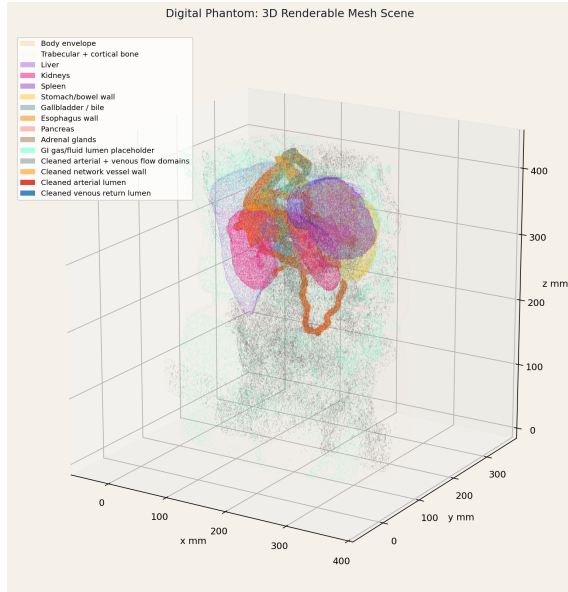


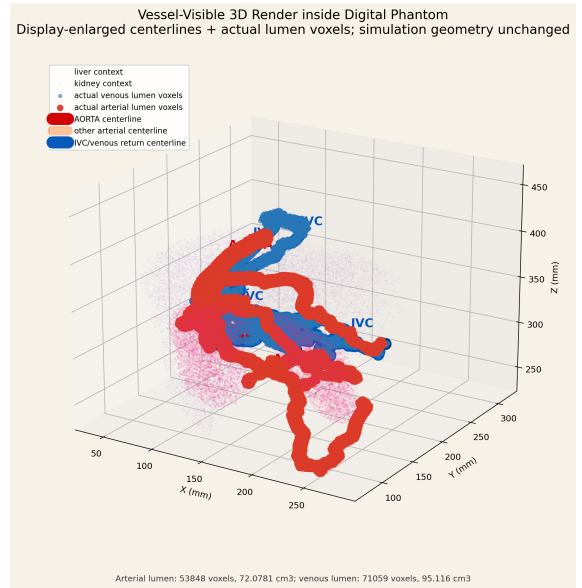
Figure 1: High-level workflow of the Adaptive Digital Phantom framework. The model combines staged imaging data, material mapping, anatomy adaptation, vascular graph modeling, flow simulation, radiotherapy QA, and renderable output generation.

4 Current Demonstrator

The current demonstrator is a torso and abdominal digital phantom with organ-preserving material maps, vascular flow domains, 3D render outputs, radiotherapy QA scaffolds, and patient/profile adaptation tools. The project is not a single static phantom; it is a framework for generating, adapting, and validating phantom states.



(a) Transparent full 3D phantom preview.



(b) Vessel-visible render mode.

Figure 2: Current 3D render outputs from the active digital phantom baseline. These figures show the anatomical context and the dedicated vessel-visible rendering mode used to inspect vascular placement.

5 Development History

Development progressed through a sequence of research milestones:

1. A project scope was defined around CT realism, radiotherapy QA, and vascular flow.
2. CT-derived torso material-label volumes were generated from staged public data.
3. Organ and vascular meshes were exported for render/CAD inspection.
4. A synthetic vascular scaffold was created with aorta, iliac, renal, hepatic, splenic, and venous pathways.
5. The vascular scaffold was voxelized into the digital phantom NIfTI volume.
6. Arterial and venous domain separation cleanup was added.
7. Flow boundary-condition packages were created.
8. Steady and pulsatile one-dimensional flow models were implemented.
9. The flow model was coupled more tightly to graph metadata and rendered over time.
10. Radiotherapy planning and QA handoff bundles were created.
11. Anthropometric torso morphing was added for BMI, waist, and height.
12. Statistical anatomy and PCA-style population morphing workflows were added.
13. Patient-case adapter workflows were added to score user CT inputs against existing profile libraries.
14. Vascular realism was improved using CTA-derived aorta and branch templates where available.

15. GI organ preservation, placeholder GI lumen support, and auto-segmentation staging bridges were added.
16. The repository was packaged for GitHub with CLI smoke tests, CI configuration, data-exclusion policy, and reproducibility documentation.

6 Core Model Components

6.1 Medical Image and Label Handling

The framework uses NIfTI volumes as the primary working representation for CT images, segmentation masks, material labels, density maps, relative electron-density maps, synthetic HU maps, vascular masks, and flow-domain labels. Dataset-specific label maps convert external segmentations into a common internal vocabulary.

Supported or staged label-map families include CT-ORG, BTCV abdomen, ImageTBAD, MedSeg abdominal vasculature, Healthy Total Body CTs, and GI tract labels. This label-map layer is important because public datasets differ in naming conventions, organ coverage, and label IDs.

6.2 Material and Density Model

The material model is defined in `configs/materials.yaml`. It provides approximate HU targets, density values, and relative electron-density values for soft tissue, lung, bone, liver, kidney, blood-equivalent fluid, contrast-enhanced vascular fluid, GI gas/fluid placeholders, and abdominal organ tissues. This gives the phantom a radiotherapy-oriented material basis rather than a purely visual segmentation basis.

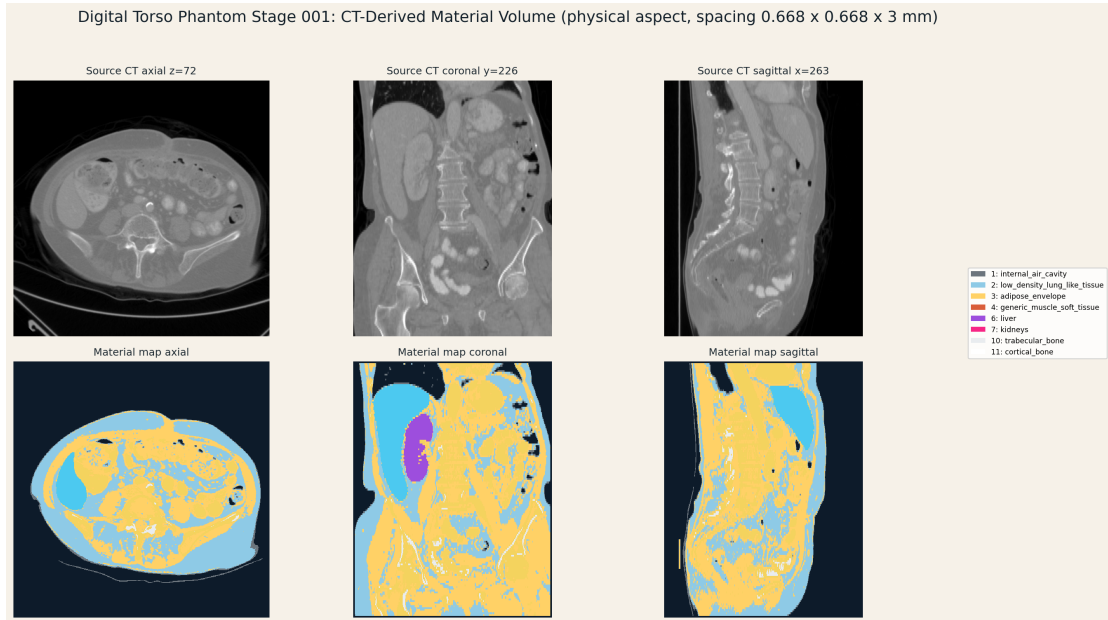


Figure 3: Example material-label torso preview from the active product baseline. Material-label maps are used to derive mass-density, relative electron-density, and synthetic-HU outputs.

6.3 Organ and GI Module

The torso module builds body masks, organ masks, material labels, mass-density maps, relative electron-density maps, and synthetic HU maps from CT and segmentation inputs. The abdom-

inal organ module preserves liver, kidneys, spleen, stomach, gallbladder, esophagus, pancreas, adrenal glands, and GI placeholder regions.

The GI module currently supports real labels where staged, placeholder/lumen layers for bowel gas or fluid, and replacement of placeholders with staged real GI masks. It also includes a bridge for supported auto-segmentation outputs.

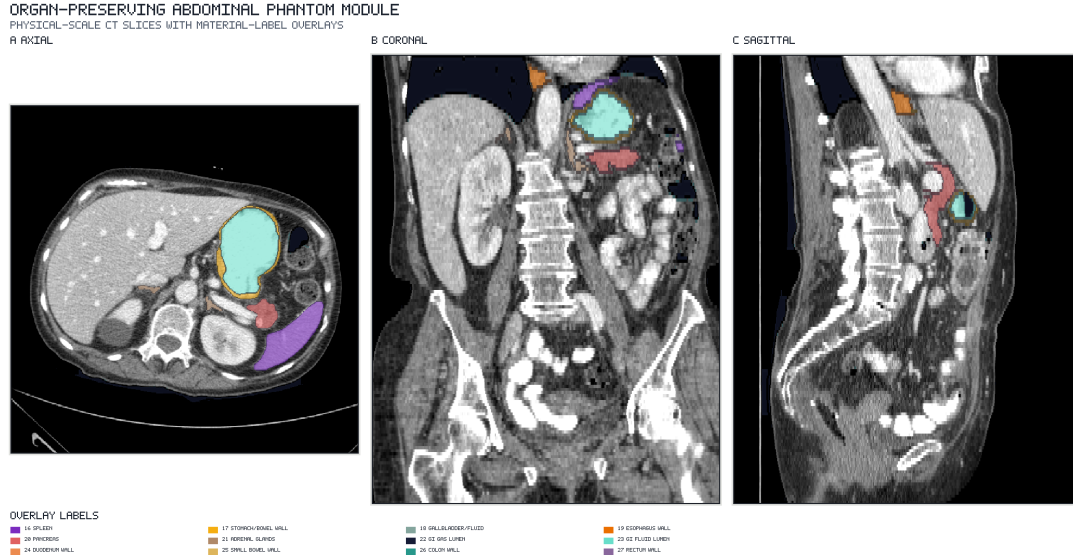


Figure 4: Abdominal organ preview showing preserved organ regions used by the product case pipeline. The GI system currently supports preserved labels and placeholder layers, with replacement by real GI masks when staged.

6.4 Anthropometric and Population Morphing

The anthropometric module adapts the torso using user-profile inputs such as BMI, waist circumference, and height. It changes the body envelope and material-label geometry in a controlled way. This provides a practical user-facing adaptation mechanism for exploring how body habitus affects anatomy, vessel placement, flow, and radiotherapy geometry.

The population cohort tools accept multiple material-label NIfTI cases and generate cohort manifests, registration-style QA, previews, PCA-ready inputs, PCA mode variants, and approved/rejected PCA mode reports. This is the pathway toward statistical anatomy deformation using real population segmentations.

6.5 Patient-Case Adapter

The patient-case adapter accepts a user CT and optional organ/vessel segmentation. It scores the input against profile and morph libraries, selects the closest adapted phantom, and can run downstream build, QA, render, flow, and release stages.

The intended workflow is:

User CT + optional organ/vessel segmentation

- > patient metrics
- > profile/morph matching
- > adapted phantom package
- > vascular deformation
- > flow simulation
- > radiotherapy QA
- > rendering and release report

7 Vascular Network and Flow Domain

The vascular model has two linked representations: a graph representation and a voxel representation. The graph representation stores nodes, edges, radii, vessel types, flow roles, and inlet/outlet identifiers. The voxel representation embeds arterial lumen, venous lumen, vessel wall, and flow-domain labels directly inside the phantom volume.

The vascular workflow supports aorta trunk generation, iliac branches, renal branches, hepatic and splenic branch placeholders, optional venous return, CTA-derived aorta trunk reconstruction where masks are available, branch-rich labeled vessel registration, bone/body collision QA, rerouting, radius-aware validation, arterial/venous separation cleanup, and flow-domain connectivity repair.

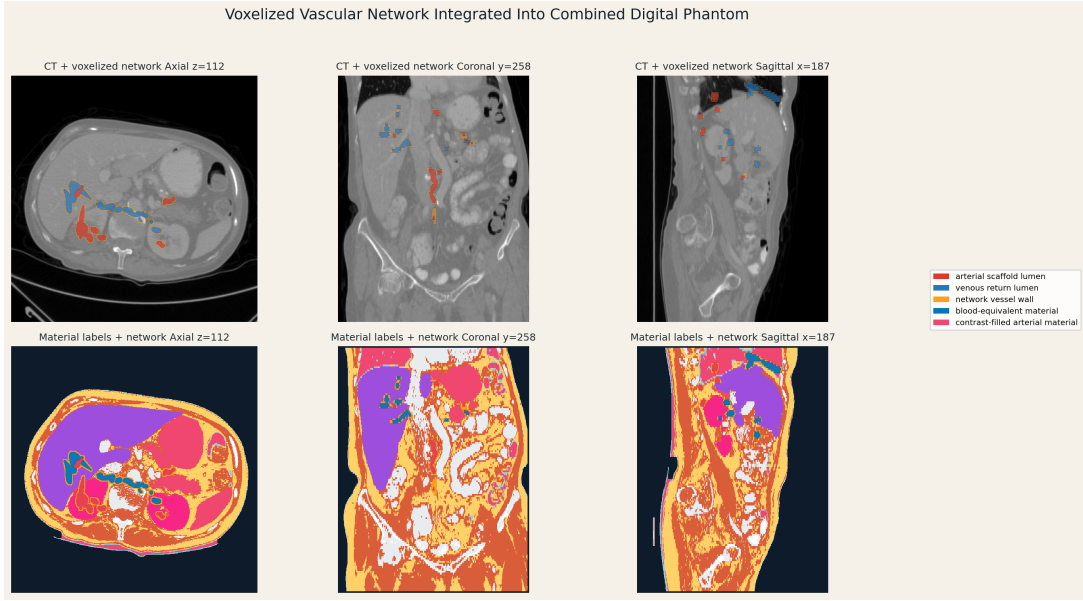
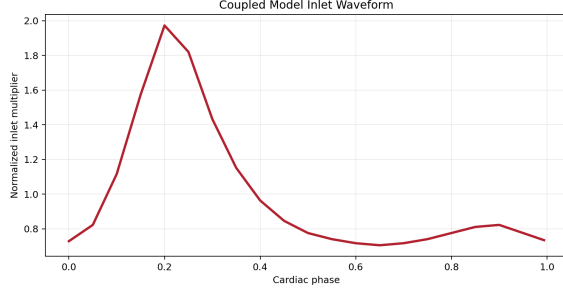


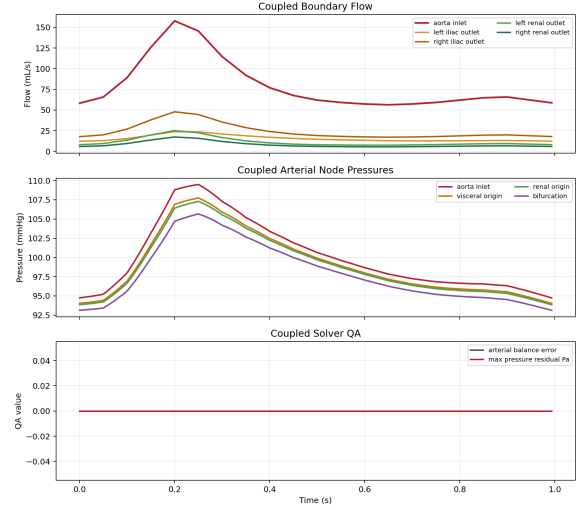
Figure 5: Voxelized vascular-domain preview. The graph-derived vascular network is embedded into the phantom volume as arterial lumen, venous lumen, vessel wall, and flow-domain labels.

8 Flow Modeling

The current flow model is a first-pass one-dimensional vascular flow simulation framework. It is intended for phantom development and QA rather than clinical hemodynamic prediction. Implemented elements include flow boundary-condition generation, inlet/outlet metadata, steady 1D flow, pulsatile flow, graph-coupled pulsatile flow, dynamic arterial outlet splitting, venous pulsatility parameters, CSV time series, and visualization-ready outputs.



(a) Pulsatile inlet waveform.



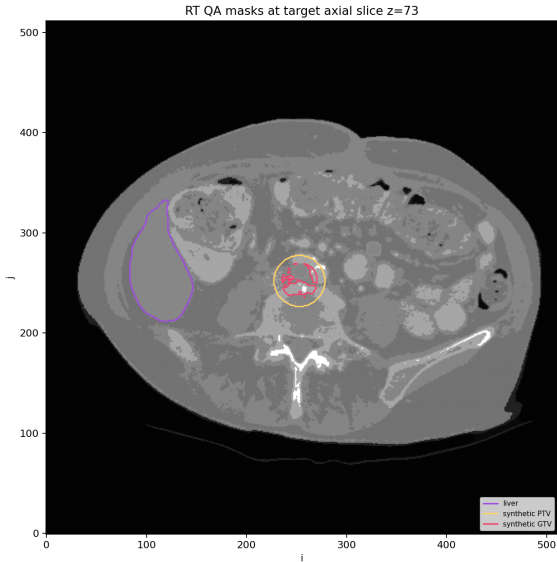
(b) Pressure/flow preview.

Figure 6: Example coupled pulsatile flow outputs generated by the active product baseline.

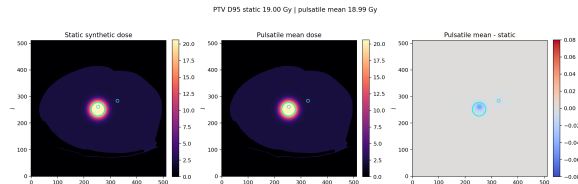
9 Radiotherapy QA

The radiotherapy module creates research handoff packages for comparing dose-related behavior across static and flow-modulated phantom states. Implemented features include RT-ready material maps, relative electron-density maps, synthetic HU maps, target and OAR-style masks, planning bundle generation, DICOM-RT-style scaffolding, static versus pulsatile dose-state comparison, PyMedPhys-compatible gamma QA support, and spatial RT-flow coupling analysis.

The current dose workflow should be interpreted as a research QA scaffold. It is not a replacement for a clinical treatment planning system or validated clinical dose engine.



(a) Radiotherapy QA preview.



(b) Planning-bundle preview.

Figure 7: Radiotherapy package and planning-bundle previews from the active product baseline.

10 Quality Assurance

The framework writes machine-readable manifests and human-readable reports at each major stage. QA modules evaluate mesh quality, vessel-organ relationships, vessel radius and bone/body conflicts, flow-domain connectivity, patient-build readiness, release readiness, and radiotherapy comparison outputs.

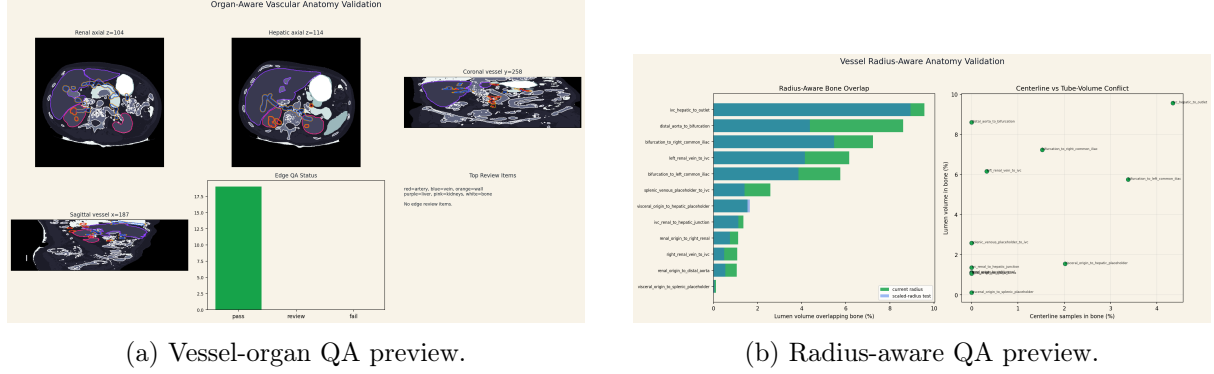


Figure 8: Example anatomical QA outputs used to evaluate vessel placement, body containment, and radius-aware conflicts.

11 Command-Line Functionality

The primary interface is `phantom-twin`. Table 1 summarizes major command groups.

Table 1: Major command groups in the `phantom-twin` CLI.

Command group	Examples and purpose
Configuration checks	<code>phase1-summary</code> , <code>materials-check</code> , <code>datasets-list</code> , <code>inspect-nifti</code> .
Mesh and rendering	<code>export-label-mesh</code> , <code>mesh-qa</code> , <code>clean-meshes</code> , <code>render-combined-3d</code> , <code>render-vascular-network-3d</code> , <code>render-3d-atlas</code> .
Torso and anatomy	<code>build-digital-torso</code> , <code>build-combined-digital-phantom</code> , <code>build-anthropometric-torso-morph</code> , <code>build-statistical-anatomy-morph</code> , <code>build-population-cohort</code> .
Patient adaptation	<code>build-patient-phantom-adapter</code> , <code>run-patient-case-adapter</code> , <code>run-patient-phantom-build</code> , <code>qa-patient-phantom-build</code> , <code>build-product-case</code> .
Vascular modeling	<code>build-vascular-network-scaffold</code> , <code>build-cta-derived-vascular-graph</code> , <code>build-labeled-vessel-vascular-graph</code> , <code>voxelize-vascular-network</code> , <code>build-label-vessel-flow-domain</code> .
Flow	<code>build-flow-boundary-package</code> , <code>build-flow-1d-model</code> , <code>build-pulsatile-flow-model</code> , <code>build-coupled-pulsatile-flow-model</code> , <code>render-4d-flow</code> .
Radiotherapy	<code>build-radiotherapy-qa-package</code> , <code>build-rt-planning-bundle</code> , <code>build-dose-gamma-qa</code> , <code>analyze-spatial-rt-flow-coupling</code> .

Command group	Examples and purpose
GI and validation	stage-gi-segmentation, auto-stage-gi-segmentation, build-validation-intake-package, harmonize-vessel-labels, qa-vessel-label-anatomy.

12 Software Dependencies

The package is defined in `pyproject.toml`. Core dependencies are `numpy` and `PyYAML`. Optional medical/research dependencies include `matplotlib`, `nibabel`, `pillow`, `pydicom`, `pymedphys`, `scikit-image`, `scipy`, `SimpleITK`, `trimesh`, `fast-simplification`, and `pytest`. The project also includes Docker and TotalSegmentator bridge scripts for GI mask staging when external segmentation tools are available.

13 Data Strategy and Reproducibility

The repository does not include medical datasets, generated NIfTI files, meshes, release archives, or patient data. Instead, it provides staging and integration pathways for public or institutionally approved data. Generated files belong in ignored directories such as `data/raw/`, `data/interim/`, `data/processed/`, `data/derived/`, `data/validation/`, and `outputs/`.

The intended clean-clone pathway is:

```
git clone https://github.com/kaiwoodger/Adaptive_Digital_Phantom-.git
cd Adaptive_Digital_Phantom-
python3.12 -m venv .venv
.venv/bin/python -m pip install --upgrade pip
.venv/bin/pip install -e ".[dev,medical]"
scripts/smoke_test_cli.sh .venv/bin/python
.venv/bin/python -m pytest -q
```

14 Practical Uses

The current framework can be used for research into patient-adaptive digital phantom workflows, anatomy/material-label pipeline testing, radiotherapy QA handoff prototyping, vascular flow-state modeling, educational 3D anatomy visualization, public segmentation benchmark development, BMI/waist/height morph comparison, vessel graph registration and collision-cleanup testing, and prototyping future physical phantom designs before printing.

15 Current Readiness

The project is best described as a publishable research and engineering demonstrator. Table 2 summarizes the current maturity.

Table 2: Current readiness of major project areas.

Area	Current status
Source-code demonstrator	High readiness; pushed to GitHub with reproducibility scaffolding.
CLI framework	High readiness for research workflows and staged local data.
Digital phantom pipeline	Moderate to high for staged cases.
Patient adaptation	Moderate; BMI/waist/height and profile matching exist but require more validation.
Vascular realism	Improving; depends on real CTA/CTV vessel masks for branch-rich realism.
GI realism	Partial; preserved organs and placeholders exist, but real bowel/colon masks are still needed.
Radiotherapy QA	Research scaffold; not TPS-equivalent.
Clinical validation	Early stage; not clinical-use ready.

16 Limitations

The model is not clinically validated. Some vascular branches remain scaffold-driven unless real CTA/CTV segmentations are staged. GI tract realism depends on the availability of real bowel, colon, rectum, and small-intestine segmentations. Anthropometric morphing is not yet a fully validated population deformation model. Flow modeling is simplified and one-dimensional. Radiotherapy dose workflows are research QA scaffolds, not replacements for clinical dose engines. Dataset access and redistribution restrictions must be handled by the user. Generated data and meshes are intentionally excluded from the repository.

17 Future Development

The strongest future development steps are:

- Stage more paired CT, CTA, CTV, organ, vessel, and GI segmentations.
- Strengthen deformable registration between patient CT anatomy and vascular templates.
- Replace remaining synthetic vascular branches with real CTA/CTV-derived branch networks.
- Add stronger organ-volume, centroid, vessel-organ, and body-shape validation.
- Improve GI segmentation support using real bowel and colon masks.
- Integrate a more realistic dose engine or treatment-planning-style workflow.
- Add richer examples using small synthetic fixtures.
- Create versioned release packages with figures, reports, and explicit limitations.
- Add optional Docker workflows for segmentation and reproducible medical-imaging dependencies.

18 Conclusion

Adaptive Digital Phantom provides a modular, reproducible foundation for building anatomically adaptable digital phantoms that combine CT-derived anatomy, material-density representation, vascular graphs, flow simulation, radiotherapy QA, rendering, and patient-profile adaptation. Its main contribution is not a single fixed phantom, but a workflow for generating, adapting, validating, and reporting digital phantom states from staged data. The project is already useful as a research demonstrator and development platform. Its next major leap in realism depends on staging more real paired CT/CTA/CTV segmentation data, strengthening deformable registration, and replacing remaining synthetic placeholders with real population-derived anatomy.

Data and Ethics Note

References